

Predicting Marathon Race Performance Using Machine Learning

Project Progress Update #1

IECE 465: Introduction to Machine Learning for Engineers

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AGENDA

1. Project Overview
2. Dataset Description
3. Data Preprocessing
4. Model Implementation
5. Preliminary Results
6. Live Demonstration
7. Future Work



PROJECT MOTIVATION

Q: Why Predict Marathon Finish Times?

Marathon performance depends on many factors including:

- Age
- Gender
- Pacing strategy
- Physical endurance

Goal of this Project:

- Predict a runner's **net marathon finish time**
- Compare performance of:
 1. Linear Regression (baseline)
 2. Neural Network Regression

Applications:

- Sports analytics
- Training optimization
- Performance prediction



DATASET OVERVIEW

Dataset: 2023 Boston Marathon

Source: Boston Marathon Results Dataset

Dataset Size:

- 26,598 runners
- 15 total variables

Key Features Used:

- *age_group*
- *gender*
- *half_time_sec*
- *finish_net_sec* (target variable)

Example Variables Available:

- Overall placement
- Gender placement
- Division placement
- Half-marathon split time
- Gun time vs. net time



DATASET PREPROCESSING

Data Cleaning Steps

1. Imported dataset into MATLAB:

```
data = readtable('boston_marathon_2023.csv');
```

2. Selected relevant features

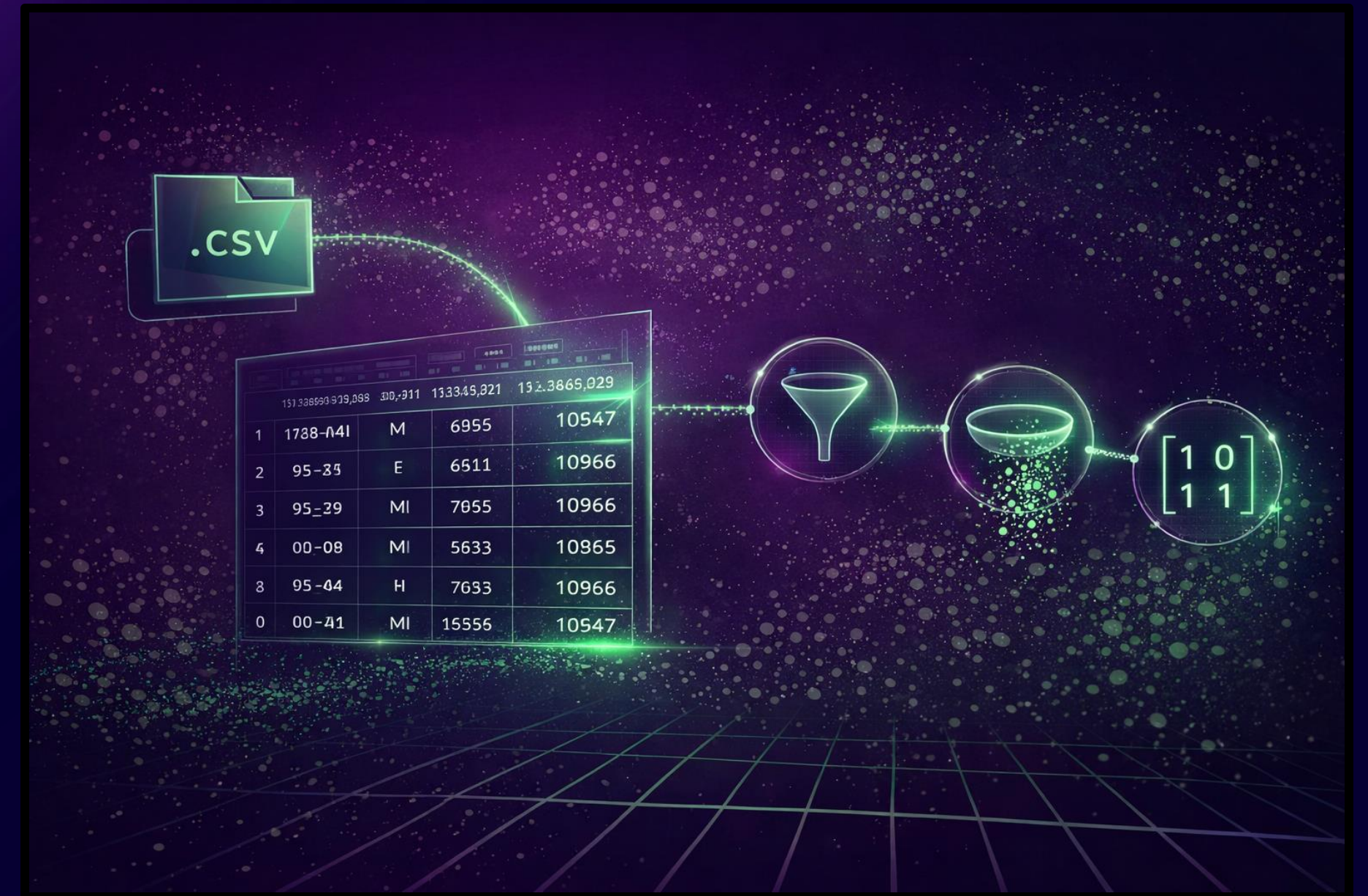
3. Removed missing values:

```
data = rmmissing(data);
```

4. Converted categorical variables:

```
data.age_group = categorical(data.age_group);
```

```
data.gender = categorical(data.gender);
```



DATA VISUALIZATION

Understanding the Dataset

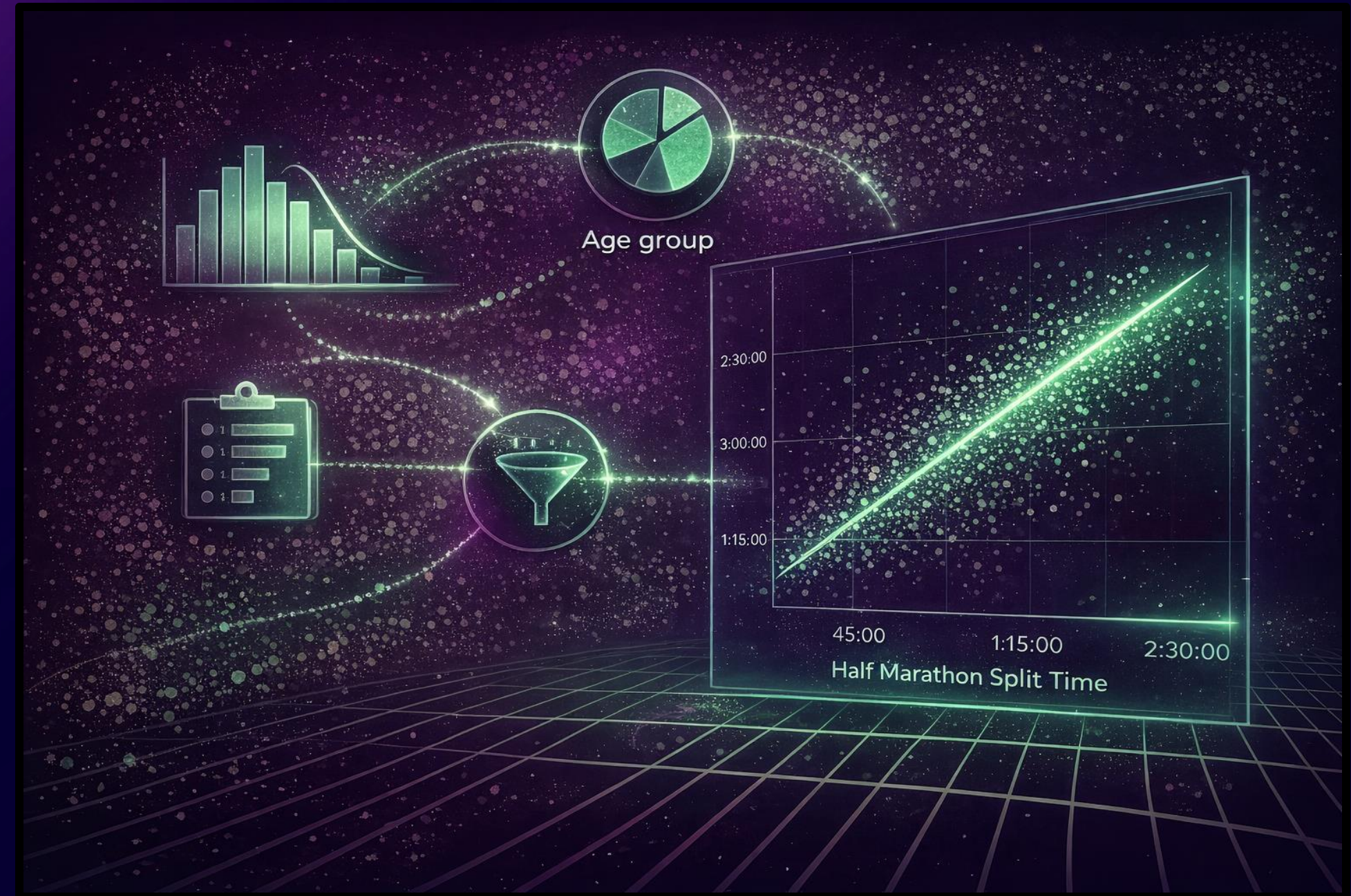
Possible Visualizations:

- Distribution of finish times
- Half marathon split vs final finish time
- Age group distribution

Key insight: Half-marathon split time is strongly correlated with final marathon time.

Suggested graph to include:

- Scatter plot
 - X-axis: half marathon time
 - Y-axis: final finish time



MODEL 1: LINEAR REGRESSION

Baseline Model

We first trained a linear regression model.

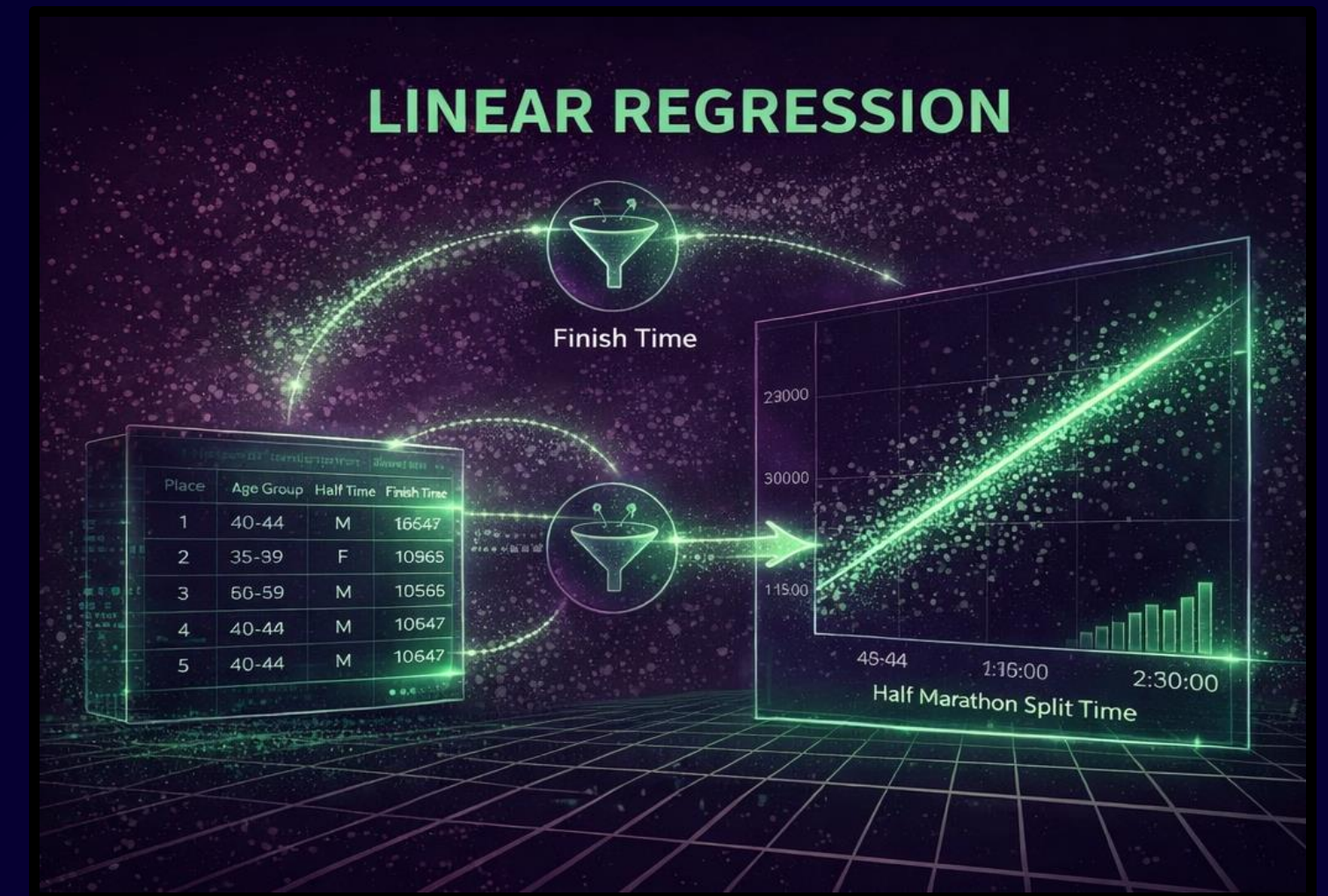
MATLAB Implementation:

```
linModel = fitlm(trainData, 'finish_net_sec ~ age_group + gender + half_time_sec');
```

Prediction: `y_pred_lin = predict(linModel, testData);`

Evaluation metrics:

- Mean absolute error (MAE)
- Root Mean Squared Error (RMSE)



MODEL 2: NEURAL NETWORK

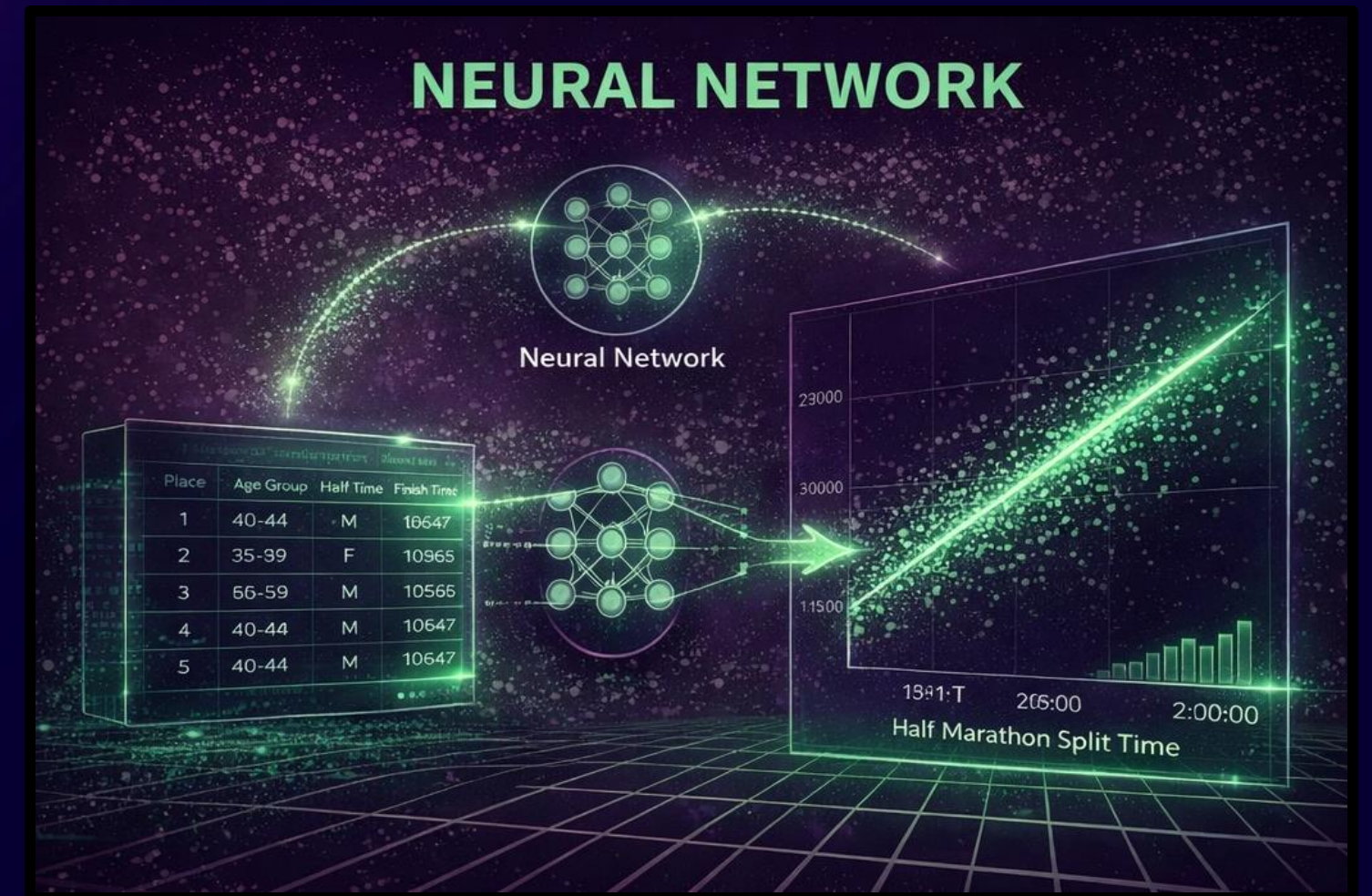
Neural Network Regression

Goal: Capture nonlinear relationships between features.

Model Architecture:

- Input features
- Hidden layers: 32 → 16 neurons
- Activation function: ReLU

MATLAB Implementation: `nnModel = fitrnet(Xtrain, ytrain, 'LayerSizes',[32 16], 'Activations','relu');`



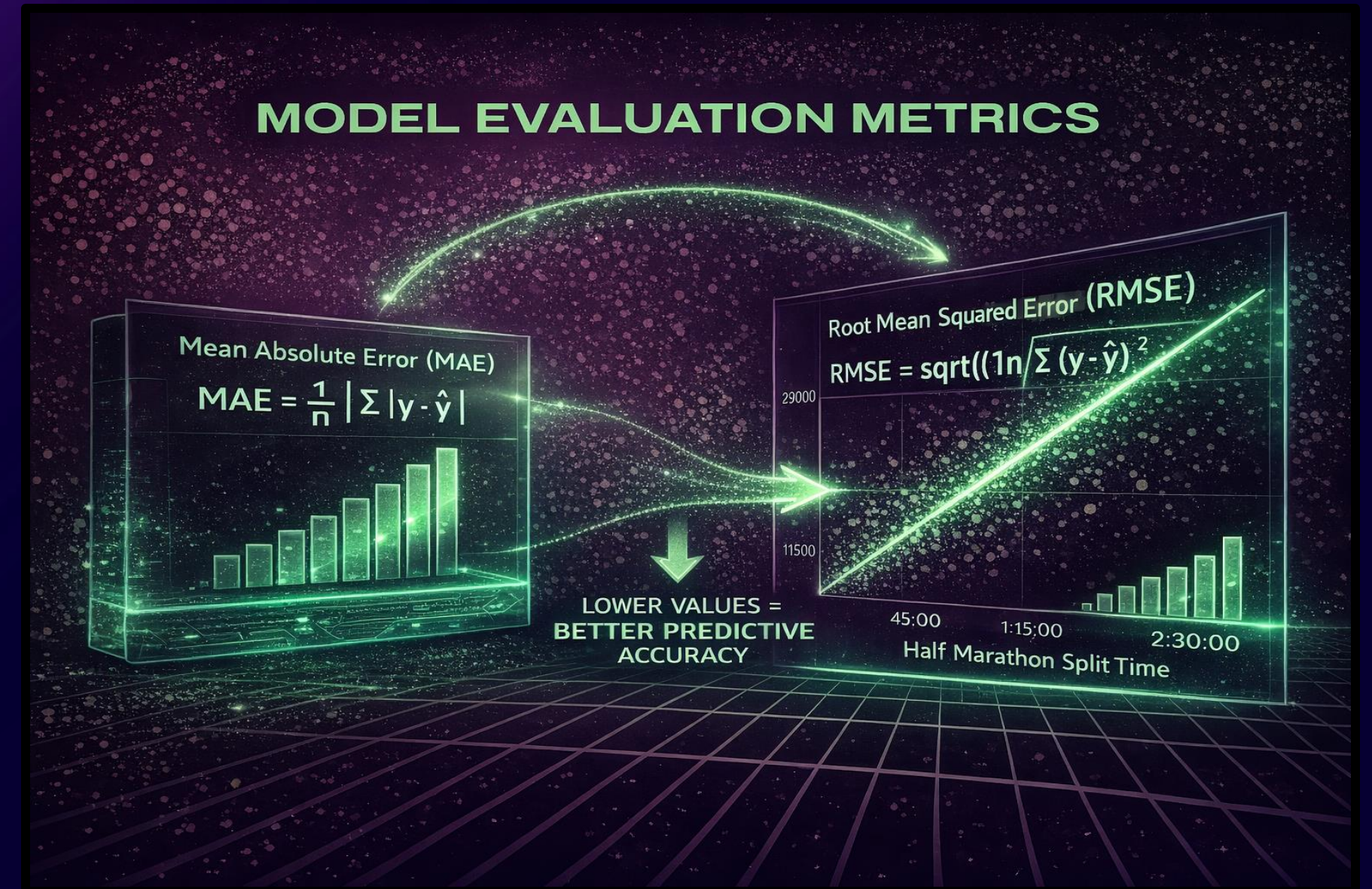
MODEL EVALUATION METRICS

Evaluation Metrics

We evaluated model performance using:

- Mean Absolute Error (MAE):
$$\text{MAE} = (1/n) * \sum |y - \hat{y}|$$
- Root Mean Squared Error (RMSE):
$$\text{RMSE} = \text{sqrt}((1/n) * \sum (y - \hat{y})^2)$$

Lower values indicate better predictive accuracy.



SYSTEM DEMONSTRATION

```
1 clear; clc; close all;
2
3 % Load dataset
4 data = readtable('boston_marathon_2023 (3).csv');
5
6 % Keep only the columns we need
7 data = data(:, {'age_group', 'gender', 'half_time_sec', 'finish_net_sec'});
8
9 % Remove rows with missing values
10 data = rmmissing(data);
11
12 % Convert categorical columns
13 data.age_group = categorical(data.age_group);
14 data.gender = categorical(data.gender);
15
16 |
17 % Train/Test Split
18 cv = cvpartition(height(data), 'HoldOut', 0.20);
19
20 trainData = data(training(cv), :);
21 testData = data(test(cv), :);
22
23
24 % MODEL 1: Linear Regression
25 linModel = fitlm(trainData, 'finish_net_sec ~ age_group + gender + half_time_sec');
26
27 y_pred_lin = predict(linModel, testData);
28 y_test = testData.finish_net_sec;
29
30 MAE_lin = mean(abs(y_test - y_pred_lin));
31 RMSE_lin = sqrt(mean((y_test - y_pred_lin).^2));
32
33 disp('----- Linear Regression Results -----')
34 disp(['MAE: ' num2str(MAE_lin)])
35 disp(['RMSE: ' num2str(RMSE_lin)])
36
37
38 % MODEL 2: Neural Network Regression
39
40 % Convert categorical vars to dummy variables
41 ageTrain = categorical(trainData.age_group);
42 genderTrain = categorical(trainData.gender);
43
44 ageTest = categorical(testData.age_group);
45 genderTest = categorical(testData.gender);
46
47 Xtrain_age = dummyvar(ageTrain);
48 Xtrain_gender = dummyvar(genderTrain);
49
50 Xtest_age = dummyvar(ageTest);
51 Xtest_gender = dummyvar(genderTest);
52
53 % Numeric feature
```


PRELIMINARY RESULTS

Model Performance Comparison:

Model	MAE (sec)	RMSE (sec)
Linear Regression	384.398	548.9473
Neural Network	381.8115	547.8768

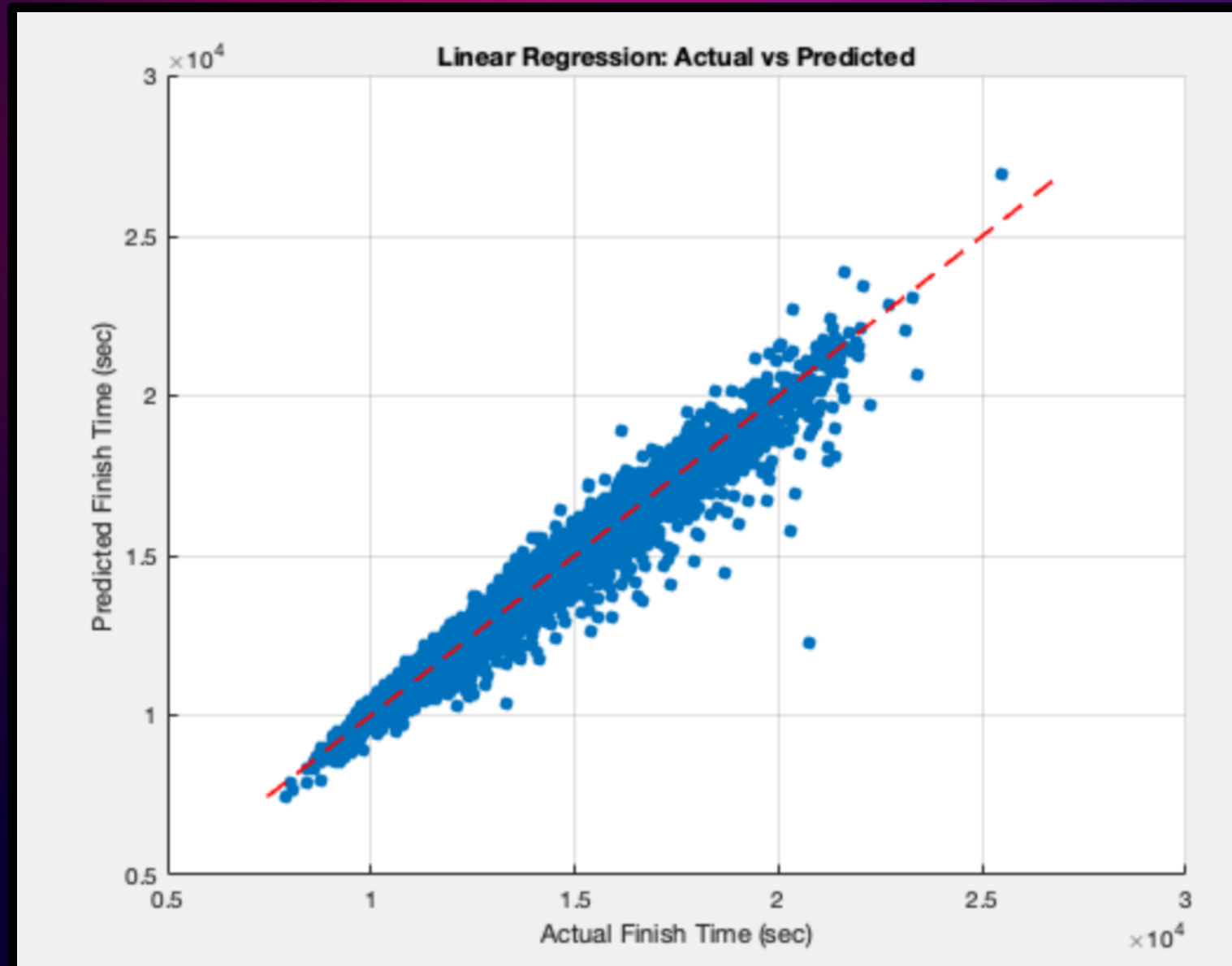
Observations:

- Neural network slightly outperforms linear regression.
- Both models perform similarly.
- Half marathon split times dominates prediction.

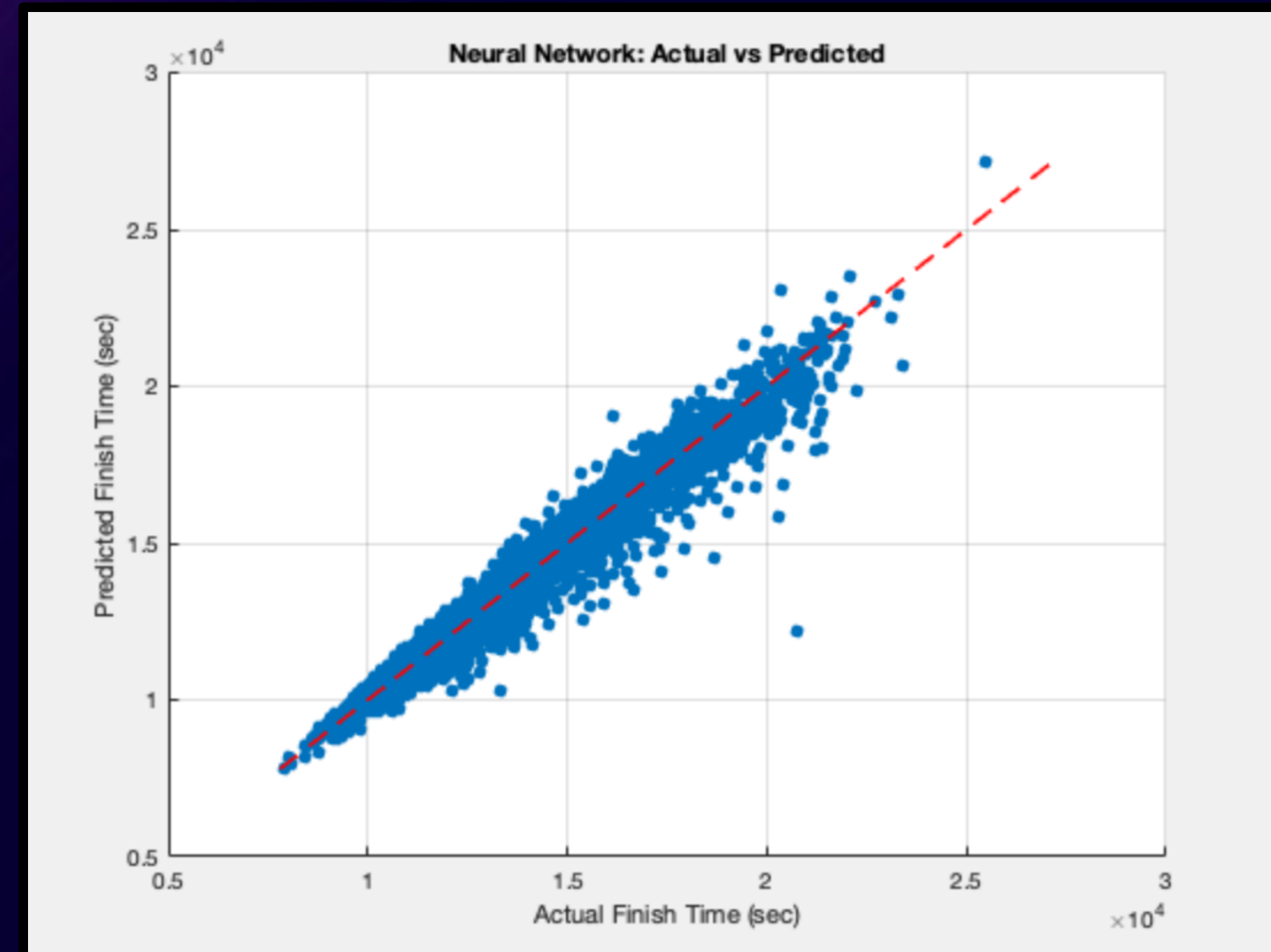
ACTUAL VS. PREDICTED RESULTS

Model Prediction Visualization:

Linear Regression Prediction Plot



Neural Network Prediction Plot



Interpretation:

- Points close to diagonal line = accurate predictions.
- Both models show strong predictive capability.

NEXT STEPS

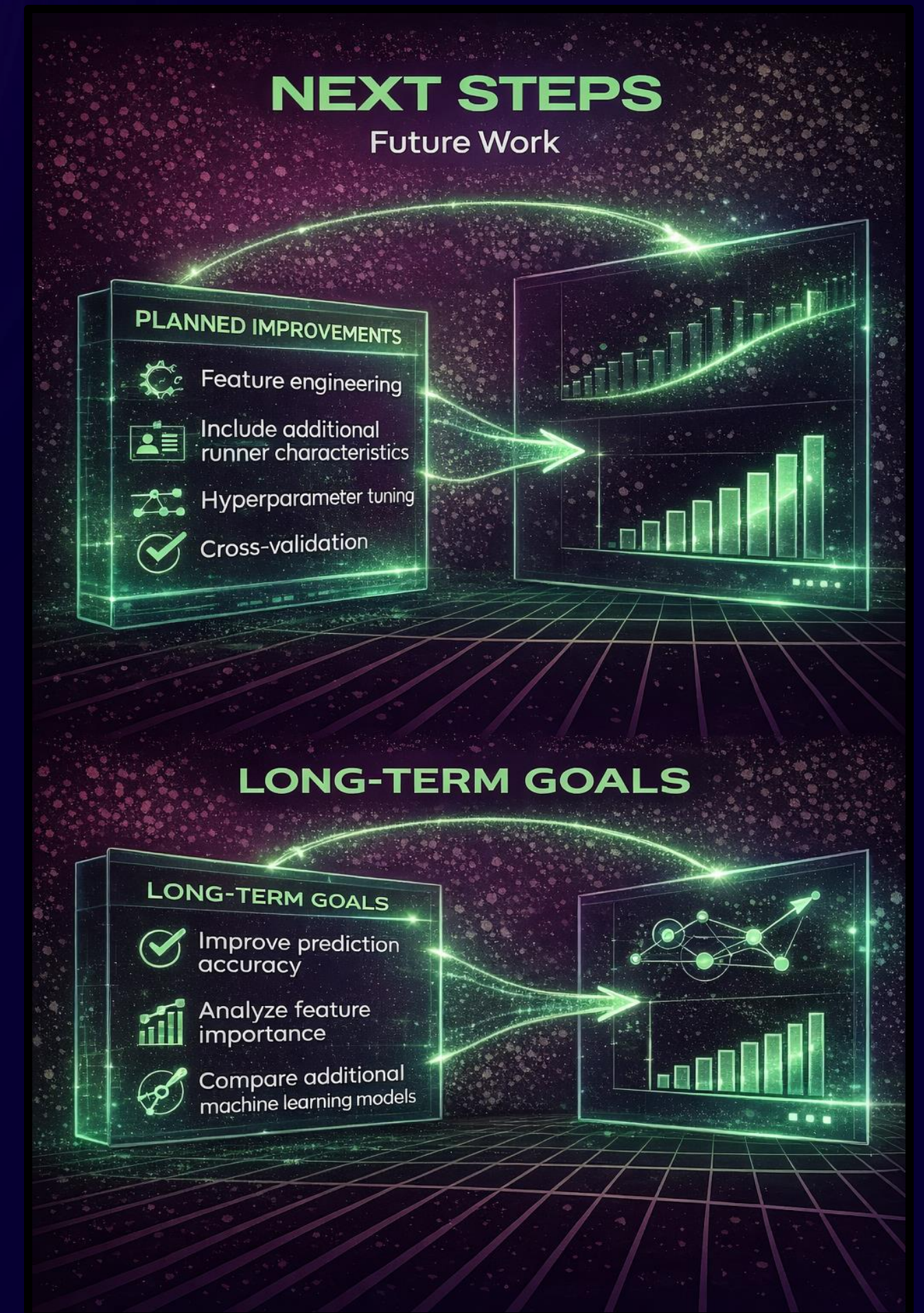
Future Work

Planned improvements:

- Feature engineering
- Include additional runner characteristics
- Hyperparameter tuning
- Cross-validation

Long-term goals:

- Improve prediction accuracy
- Analyze feature importance
- Compare additional machine learning models



Questions?

